Giacomo Enrico Sodini

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Research Interests

My main research interests lie at the intersection of Optimal Transport, Non-Smooth analysis, and Calculus of Variations. I work on Unbalanced Optimal Transport, on metric Sobolev and BV spaces on metric-measure spaces, and I study geometry and evolutions in spaces of (probability) measures.

Academic Positions

PostDoc *Dec 2025 – Present*
Institute of Analysis and Scientific Computing - TU Wien, Vienna

University Assistant *Oct 2022 – Dec 2025*
Fakultät für Mathematik - Universität Wien, Vienna

Education

Ph.D. in Mathematics *Nov 2019 – Sept 2022*
TUM-IAS, Munich
Thesis: *Optimal transport: unbalanced positive measures, dissipative evolutions and Sobolev spaces*
Supervisors: Prof. M. Fornasier and Prof. G. Savaré
Grade: *summa cum laude*

M.Sc. in Mathematical Engineering *Oct 2017 – Oct 2019*
Politecnico di Milano

B.Sc. in Mathematical Engineering *Oct 2014 – Jul 2017*
Politecnico di Milano

Publications

Preprints

[18] Massimo Fornasier and Giacomo Enrico Sodini. "Approximation in Metric Sobolev Spaces: A General Framework". 2026. 19 pp. arXiv: 2606.23282.

[17] Pierre-Cyril Aubin-Frankowski, Giacomo Enrico Sodini, and Ulisse Stefanelli. "The Brezis-Ekeland-Nayroles principle in metric spaces: time-continuous setting". 2026. 22 pp. arXiv: 2606.14432.

[16] François Delarue, Mattia Martini, and Giacomo Enrico Sodini. "HJB equations driven by the Dirichlet-Ferguson Laplacian in Wasserstein-Sobolev spaces". 2025. 66 pp. arXiv: 2511.03522.

[15] Nicolò De Ponti, Giacomo Enrico Sodini, and Luca Tamanini. "The infimal convolution structure of the Hellinger-Kantorovich distance". 2025. 53 pp. arXiv: 2503.12939.

[14] Lorenzo Dello Schiavo and Giacomo Enrico Sodini. "The Hellinger-Kantorovich metric measure geometry on spaces of measures". 2025. 96 pp. arXiv: 2503.07802.

Journal Articles

[13] Giulia Cavagnari, Giuseppe Savaré, and Giacomo Enrico Sodini. "Stochastic Euler Schemes and Dissipative Evolutions in the Space of Probability Measures". *Mathematical Models and Methods in Applied Sciences* (2026). 53 pp. DOI: 10.1142/S0218202526500521.

[12] Goro Akagi, Giacomo Enrico Sodini, and Ulisse Stefanelli. "Global well-posedness for a time-fractional doubly nonlinear equation". *Journal of Differential Equations* 474 (2026), Paper No. 114528. 28 pp. DOI: 10.1016/j.jde.2026.114528.

[11] Giulia Cavagnari, Giuseppe Savaré, and Giacomo Enrico Sodini. "A Lagrangian approach to totally dissipative evolutions in Wasserstein spaces". *Journal of Differential Equations* 470 (2026), Paper No. 114395. 127 pp. DOI: 10.1016/j.jde.2026.114395.

[10] Pierre-Cyril Aubin-Frankowski, Giacomo Enrico Sodini, and Ulisse Stefanelli. “Evolution variational inequalities with general costs”. *J. Funct. Anal.* 291.1 (2026), Paper No. 111469. 51 pp. DOI: 10.1016/j.jfa.2026.111469.

[9] Enrico Pasqualetto and Giacomo Enrico Sodini. “Functions of bounded variation and Lipschitz algebras in metric measure spaces”. *ESAIM: COCV* 32 (2026), Paper No. 28. 24 pp. DOI: 10.1051/cocv/2026014.

[8] Massimo Fornasier, Pascal Heid, and Giacomo Enrico Sodini. “Approximation Theory, Computing, and Deep Learning on the Wasserstein Space”. *Mathematical Models and Methods in Applied Sciences* 35.04 (2025), pp. 825–903. 79 pp. DOI: 10.1142/S0218202525500113.

[7] Giulia Cavagnari, Giuseppe Savaré, and Giacomo Enrico Sodini. “Extension of monotone operators and Lipschitz maps invariant for a group of isometries”. *Canad. J. Math.* 77.1 (2025), pp. 149–186. 38 pp. DOI: 10.4153/S0008414X23000846.

[6] Giuseppe Savaré and Giacomo Enrico Sodini. “A relaxation viewpoint to unbalanced optimal transport: duality, optimality and Monge formulation”. *J. Math. Pures Appl.* 188 (9 2024), pp. 114–178. 65 pp. DOI: 10.1016/j.matpur.2024.05.009.

[5] Giacomo Enrico Sodini. “The general class of Wasserstein Sobolev spaces: density of cylinder functions, reflexivity, uniform convexity and Clarkson’s inequalities”. *Calc. Var. Partial Differential Equations* 62.7 (2023), Paper No. 212. 41 pp. DOI: 10.1007/s00526-023-02543-1.

[4] Massimo Fornasier, Giuseppe Savaré, and Giacomo Enrico Sodini. “Density of subalgebras of Lipschitz functions in metric Sobolev spaces and applications to Wasserstein Sobolev spaces”. *J. Funct. Anal.* 285.11 (2023), Paper No. 110153. 76 pp. DOI: 10.1016/j.jfa.2023.110153.

[3] Giulia Cavagnari, Giuseppe Savaré, and Giacomo Enrico Sodini. “Dissipative probability vector fields and generation of evolution semigroups in Wasserstein spaces”. *Probab. Theory Related Fields* 185.3-4 (2023), pp. 1087–1182. 96 pp. DOI: 10.1007/s00440-022-01148-7.

[2] Giuseppe Savaré and Giacomo Enrico Sodini. “A simple relaxation approach to duality for optimal transport problems in completely regular spaces”. *J. Convex Anal.* 29.1 (2022), pp. 1–12. 12 pp.

[1] Mattia Martini and Giacomo Enrico Sodini. “Numerical methods for a system of coupled Cahn-Hilliard equations”. *Commun. Appl. Ind. Math.* 12.1 (2021), pp. 1–12. 12 pp. DOI: 10.2478/caim-2021-0001.

Books

[B1] Mauro D’Amico et al. “Mathematical Analysis - Module 1 Exercises”. Vol. 1. BAI Series. Università Bocconi, EGEA, 2021. 176 pp.

Theses

[T1] Giacomo Enrico Sodini. “Optimal Transport: unbalanced positive measures, dissipative evolutions and Sobolev spaces”. PhD thesis. Technische Universität München, 2022. 296 pp.

Talks

Invited Seminars

- **06.2026** Faculty of Mathematics, Informatics and Mechanics, University of Warsaw, Warsaw
- **03.2026** Department of Mathematics, Politecnico di Milano, Milano
- **11.2025** Department of Mathematics, University of Naples Federico II, Naples
- **11.2025** Institut für Statistik, TU Graz, Graz
- **10.2025** VADOR group, Institute of Analysis and Scientific Computing, TU Vienna, Vienna
- **10.2025** Department of Decision Sciences, Bocconi University, Milano
- **04.2025** Applied Mathematics and Modeling group, Faculty of Mathematics, University of Vienna, Vienna
- **11.2024** Department of Mathematics, University of Jyväskylä, Jyväskylä
- **10.2024** Department of Mathematical Sciences, University of Durham, Durham
- **11.2023** Laboratoire J.A. Dieudonné, Université Côte d’Azur, Nice
- **11.2023** Mathematical Finance and Probability Theory group, Faculty of Mathematics, University of Vienna, Vienna

- **05.2023** PDE Afternoon, University of Vienna, Vienna
- **01.2023** Stochastic Analysis Group, ISTA, Klosterneuburg
- **10.2022** Applied Mathematics and Modeling group, Faculty of Mathematics, University of Vienna, Vienna
- **07.2022** KU-LMU-TUM Joint Seminar, TU Munich, Munich
- **05.2022** Research Group Data Science, TU Munich, Munich
- **11.2021** Department of Mathematics, Politecnico di Milano, Milano
- **04.2020** SeMiNarri di Matematica, University of Pavia, Pavia

Invited Conference Talks

- **09.2025** *The Annual 2025 ÖMG-DMV Meeting*, JKU, Linz
- **05.2024** *Lions-Magenes days 2024*, University of Pavia, Pavia
- **04.2024** *Variational Analysis, Models and Methods in Measure Spaces*, CIRM, Marseille
- **09.2023** *The Mathematics of Subjective Probability*, Bicocca University, Milano
- **04.2023** *OTMFML*, IAS/TUM, Munich
- **11.2022** *Geometric Analysis and PDEs at PoliMi*, Politecnico di Milano, Milano
- **11.2022** *Smooth Functions on Rough Spaces and Fractals with Connections to Curvature Functional Inequalities*, BIRS, Banff
- **10.2022** *Optimal Transportation and Application*, SNS, Pisa

Contributed Talks

- **01.2026** *Dolomites Winter School*, Folgarida
- **10.2025** *Austrian Calculus of Variations day*, University of Graz, Graz
- **01.2025** *Dolomites Winter School*, Folgarida
- **11.2024** *Austrian Calculus of Variations day*, University of Innsbruck, Innsbruck
- **09.2024** *MeRiOT*, Varenna
- **11.2023** *Austrian Calculus of Variations day*, TU Wien, Vienna
- **11.2022** *Austrian Calculus of Variations day*, University of Salzburg, Salzburg
- **06.2022** *PIMS-IFDS-NSF Summer School on Optimal Transport*, University of Washington, Seattle

Teaching

Lecturer

- **TU Wien**
  - Introduction to Optimal Transport (SoSe 2026)
- **University of Vienna**
  - Topics in the Calculus of Variations (SoSe 2025)

Teaching Assistant

- **University of Vienna**
  - Introduction to Analysis (WiSe 2025-2026)
  - Analysis 3 (SoSe 2025)
  - Topologie und Funktionalanalysis (WiSe 2024-2025, WiSe 2023-2024)
  - Analysis 2 (SoSe 2024)
- **Bocconi University**
  - Mathematical Analysis - Module 1 (2025-2026, 2024-2025, 2023-2024, 2022-2023, 2021-2022, 2020-2021)
- **TU Munich**
  - Foundations in Data Analysis (SoSe 2022, SoSe 2021, SoSe 2020)
- **Politecnico di Milano**
  - Mathematical Analysis II (2021-2022, 2020-2021)

Service

Organization

- **06.2026** Contributed session - The 5th Italian Meeting on Probability and Mathematical Statistics, University of Palermo, Palermo

Peer Review

Discrete and Continuous Dynamical Systems; Annali della Scuola Normale Superiore di Pisa - Classe di Scienze; ESAIM: Control, Optimisation and Calculus of Variations; Journal of Functional Analysis; Networks and Heterogeneous Media; SIAM Journal on Mathematics of Data Science; Canadian Mathematical Bulletin; SIAM Journal on Mathematical Analysis; Journal of Evolution Equations.

Thesis Supervision

- Lea Laimer, *Weak topologies and the James space* (Bachelor thesis, University of Vienna)

Last updated: August 21, 2026